Retiming

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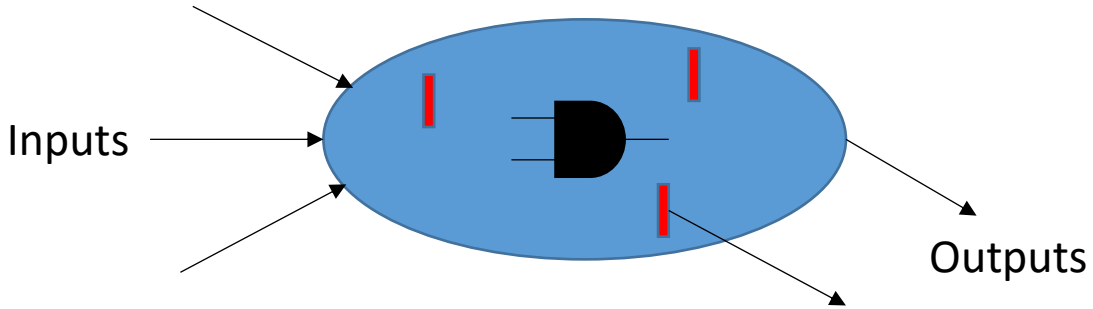
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Outline

- Retiming Problem Formulation
- Retiming Algorithm

Optimizing Sequential Circuits by Retiming

- Netlist of gates and registers:



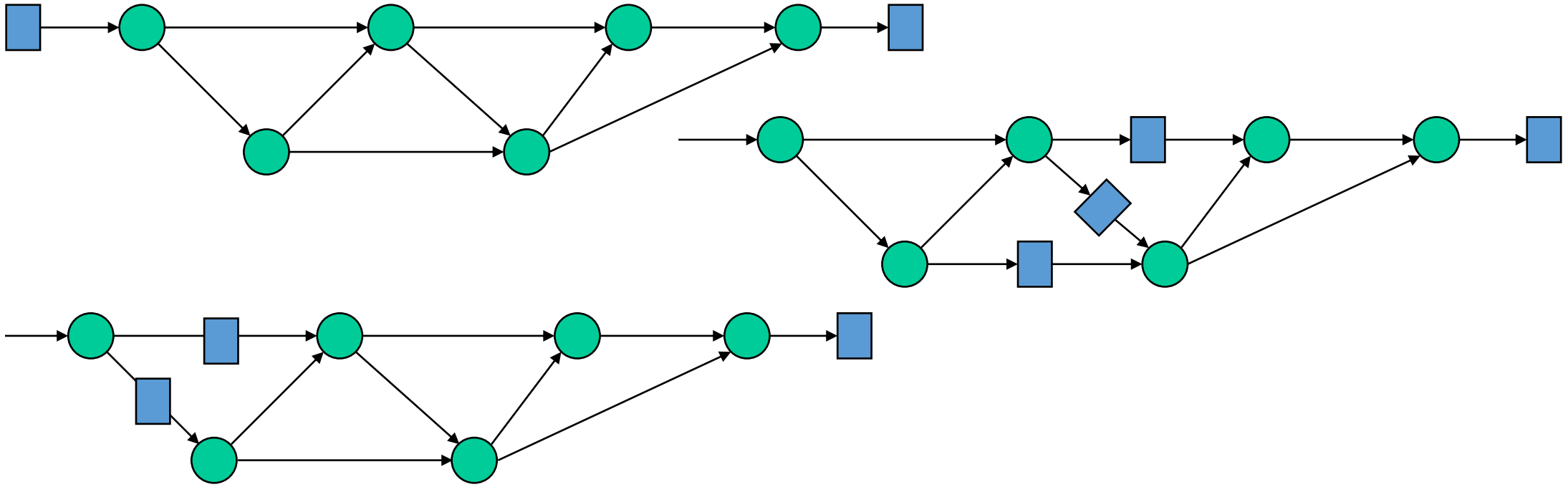
- Various Goals:
  - Reduce clock cycle time
  - Reduce area
    - Reduce number of registers

Retiming

- Problem
  - Pure combinational optimization can be myopic since relations across register boundaries are disregarded.
- Solution
  - Retiming: Move register (s) so that
    - Clock cycle decreases, or number of registers decreases and
    - Input-output behavior is preserved
  - Peripheral retiming: Combine retiming with combinational optimization techniques
    - Move latches out of the way temporarily
    - Optimize larger blocks of combinational logic

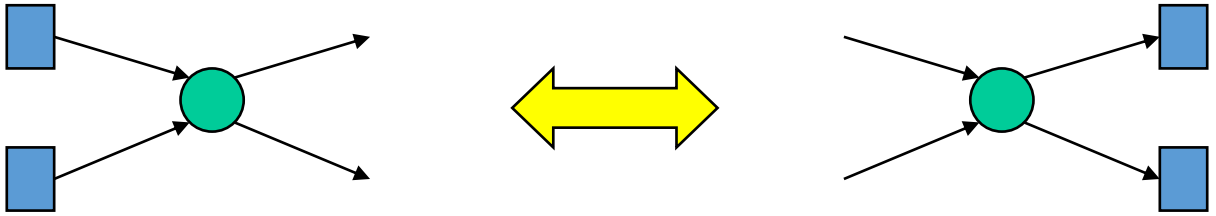
Retiming

- Process of optimally distributing registers throughout a circuit
  - minimize the clock period
  - minimize the number of registers



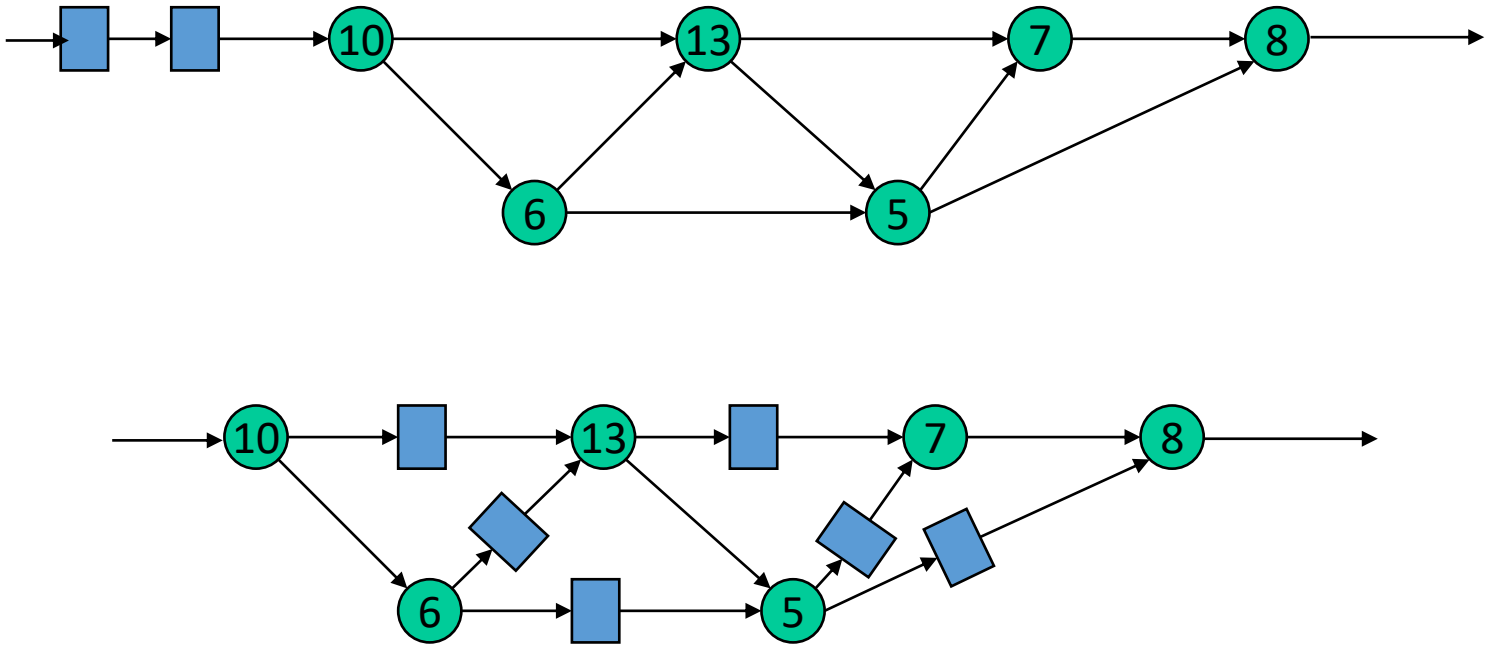
Retiming (cont.)

- Retiming rules:
  - remove one register from each input and add one to each output
  - remove one register from each output and add one to each input



Optimal Pipelining

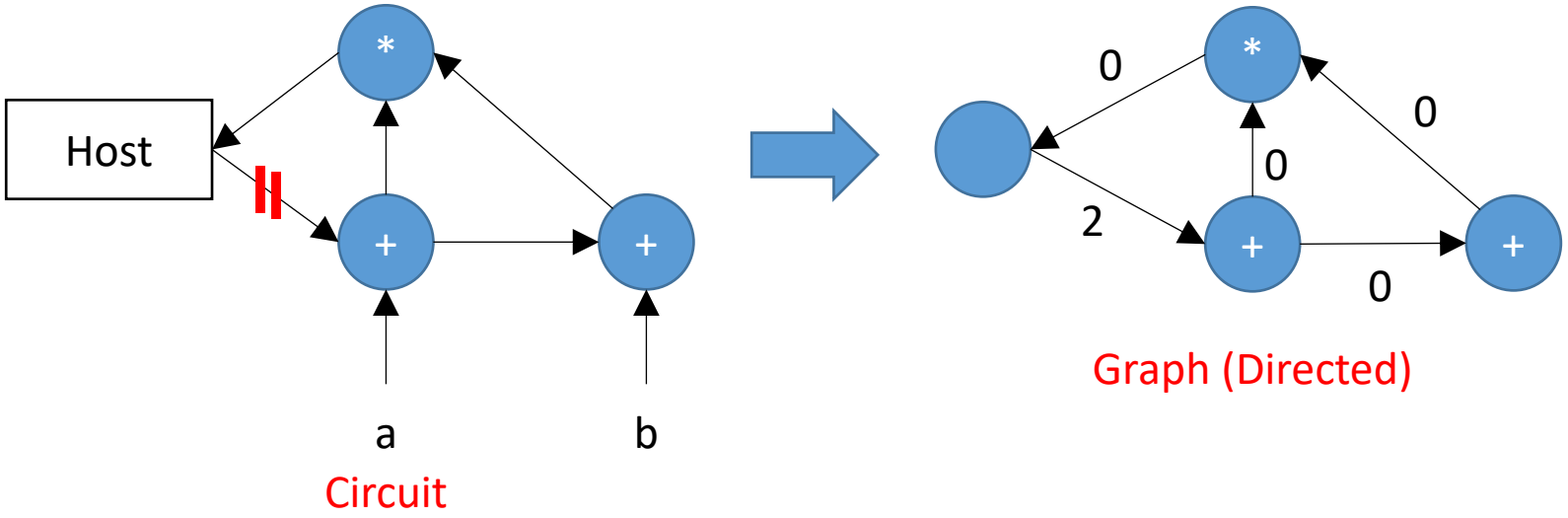
- Add registers - use retiming to find optimal location



Circuit Representation

- Circuit representation: G (V, E, d, w)
  - V $\leftrightarrow$ set of gates
  - E $\leftrightarrow$ set of wires
  - d(v) = delay of gate/vertex v, $(d(v) \geq 0)$
  - w(e) = number of registers on edge e, $(w(e) \geq 0)$

Circuit Representation



- Every cycle in the graph has at least one register, i.e. there are no combinational loops

Operation	Delay
+	3
*	7

Preliminaries

- For a path p: $V_0 \xrightarrow{e_0} V_1 \xrightarrow{e_1} \ldots V_{k-1} \xrightarrow{e_{k-1}} V_k$

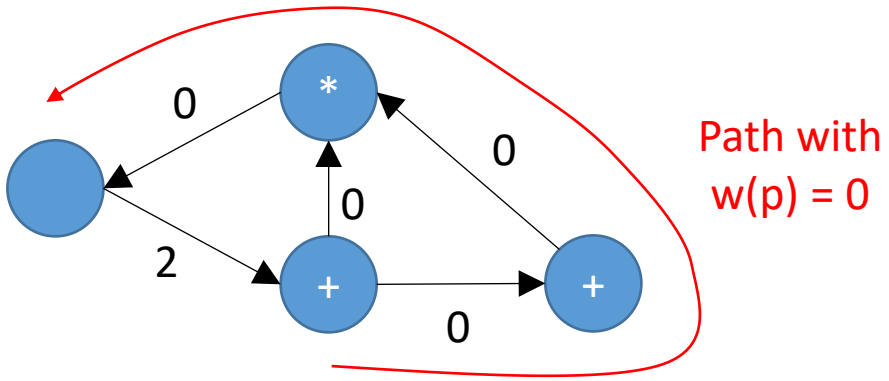
$$d(p) = \sum_{i=0}^{k} d(V_i) \ (includes\ end\ points)$$

$$w(p) = \sum_{i=0}^{k-1} w(e_i)$$

- Clock cycle C:

$$C = \max_{p:w(p)=0} \{d(p)\}$$

- For the correlator circuit: *c=13*
- Can we reduce it to 7? How?



Preliminaries: Solving Inequalities

- Given a system of M inequalities in N variables.
- Each inequalities has the form $r_i - r_j <= k$, k is an integer.
- Shortest path algorithm can be used to solve them.
- Ex1. An example with M=5 and N=4 is given.

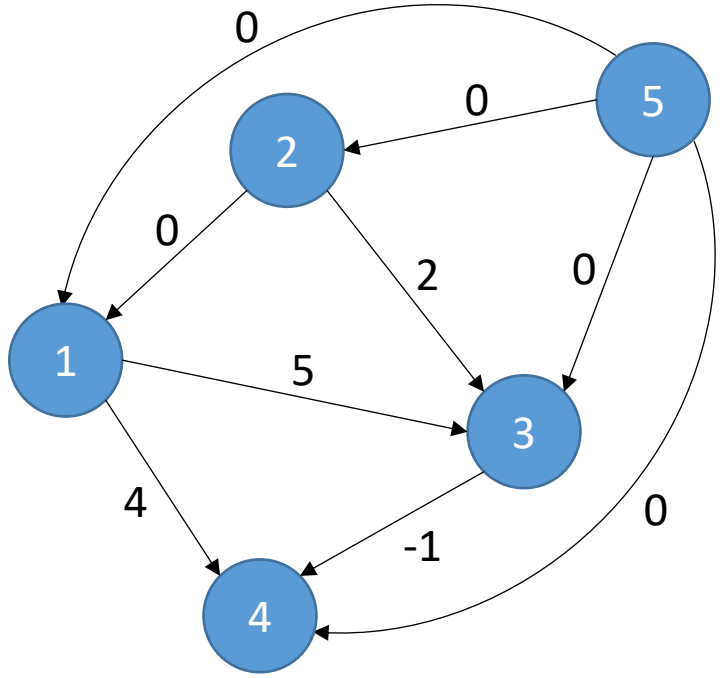
$$r_1 - r_2 \leq 0$$
$$r_3 - r_1 \leq 5$$
$$r_4 - r_1 \leq 4$$
$$r_4 - r_3 \leq -1$$
$$r_3 - r_2 \leq 2$$

Preliminaries: Constraint Graph

- Inequalities can be solved using constraint graph.
- Drawing a constraint Graph:
  a. Draw the node i for each of the N variables r_i, i = 1, 2, 3, … , N.
  b. Draw node N + 1.
  c. For each inequality $r_i - r_j <= k$, draw the edge j->i, from j to i of length k.
  d. For each node i, i = 1, 2, 3, … , N, draw the edge N+1 -> i from node N+1 to i with length 0.

Preliminaries: Constraint Graph

$$r_1 - r_2 \leq 0$$
$$r_3 - r_1 \leq 5$$
$$r_4 - r_1 \leq 4$$
$$r_4 - r_3 \leq -1$$
$$r_3 - r_2 \leq 2$$



r1 = 0, r2 = 0, r3 =0, r4 = -1, and r5 = 0

Preliminaries: Solve Using Bellman-Ford Algorithm

- The system of inequalities has a solution if and only if the constraint graph contains no negative cycles.
- If a solution exists, one solution is where r_i is the minimum length path from the node N+1 to node i.
- Origin node in the fig.1 is node 5.
- Using Bellman-Ford algorithm we find that there are no negative cycles.

Preliminaries: Solve Using Bellman-Ford Algorithm

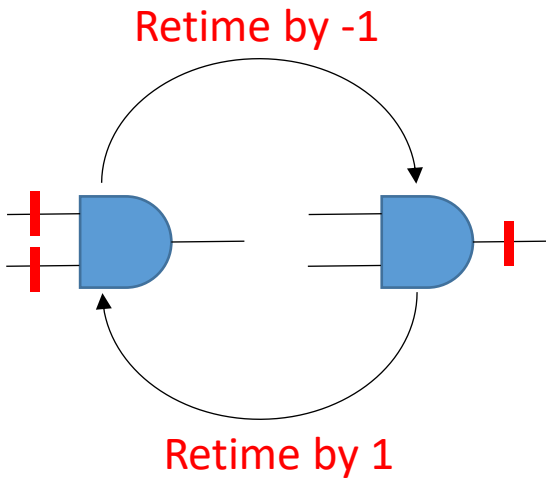
- Solution can be found by examining $r^{(4)}(V)$, V= 1, 2, 3, 4.
- Shortest distance of each node from source 5 is, $r^{(4)}(1) = 0$, $r^{(4)}(2) = 0$, $r^{(4)}(3) = 0$, $r^{(4)}(4) = -1$, and $r^{(4)}(5) = 0$.
- Therefore a solution to the system of inequalities is determined to be $r_1=0$, $r_2=0$, $r_3=0$, and $r_4= -1$.

Preliminaries: Using Floyd-Warshall Algorithm

$$R^{(6)} = \begin{bmatrix} \infty & \infty & 5 & 4 & \infty \\ 0 & \infty & 2 & 1 & \infty \\ \infty & \infty & \infty & -1 & \infty \\ \infty & \infty & \infty & \infty & \infty \\ 0 & 0 & 0 & -1 & \infty \end{bmatrix}$$

Basic Operation

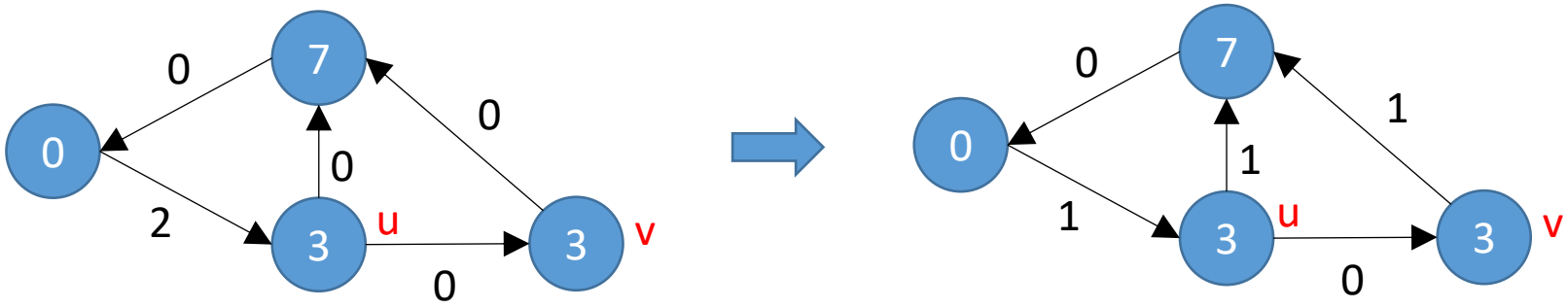
- Movement of registers
  - from input to output of a gate or vice versa
- Does not affect gate functionalities
- Vertex labeling
  - $r: V \rightarrow Z, an\ integer\ vertex\ labeling$
  - $w_r(e) = w(e) + r(v) - r(u) for\ edge\ e = (u, v)$





Basic Operation

- In the example, $r(u) = -1, r(v) = -1$ results in



- For a path $p: s \to t, w_r(p) = w(p) + r(t) - r(s)$
- Retardation
  - $r: V \to Z$ an integer vertex labeling
  - $w_r(e) = w(e) + r(v) - r(u)$ for edge $e = (u, v)$
  - A retiming r is legal if $w_r(e) \geq 0, \forall e \in E$

Retiming for Minimum Clock Cycle

- **Problem Statement: (minimum cycle time)**

  Given $G(V, E, d, w)$, find a legal retiming *r* so that

  $$C = \max_{p:w(p)=0} \{d(p)\}$$

  is minimized

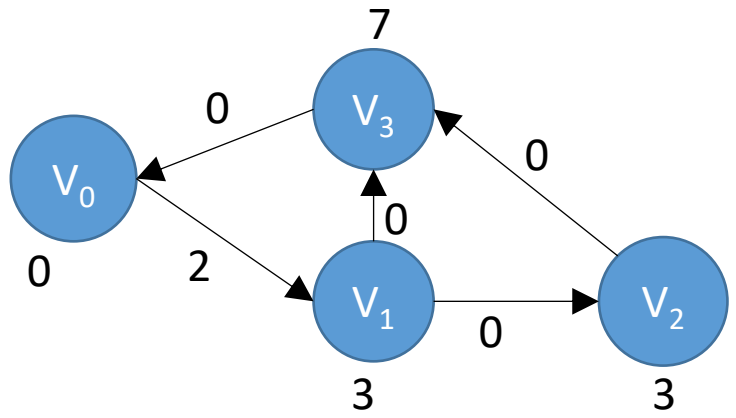
- **Retiming: 2 important matrices**
  - Register weight matrix

    $$\mathrm{W}(u, v) = \min_{p}\{w(p): u \xrightarrow{p} v\}$$

  - Delay matrix

    $$D(u, v) = \max_{p}\{d(p): u \xrightarrow{p} v, w(p) = W(u, v)\}$$

Retiming for Minimum Clock Cycle



W=register path weight matrix
(minimum # latches on all paths between u and v)
D= path delay matrix
(maximum delay on all paths between u and v with minimum number of latches)

$$c \leq \propto \Leftrightarrow \forall p, if\ D(p) > \alpha\ then\ W(p) \geq 1$$

W

	V_0	V_1	V_2	V_3
V_0	0	2	2	2
V_1	0	0	0	0
V_2	0	2	0	0
V_3	0	2	2	0

D

V_0	V_1	V_2	V_3	
0	3	6	13	V_0
13	3	6	13	V_1
10	13	3	10	V_2
7	10	13	7	V_3

Delays exceeding 7 shown in red

Conditions for Legal Retiming

- Assume that we are asked to check if a retiming exists for a clock cycle α

(1) Feasibility Constraint:

- $w_r(e) \geq 0 \; for \; all \; e$. That is the number of registers in each edge cannot be negative.
- $w_r(e) = w(e) + r(v) - r(u) \geq 0 \; or$
- $r(u) - r(v) \leq w(e)$

(2) Critical path constraint:

r(u) – r(v) <= W(u, v) – 1 for all vertices u and v in G such that D(u, v) > α

Conditions for Legal Retiming

- For all paths $p: u \rightarrow v$ such that $d(p) > \propto$, we require $w_r(p) \geq 1$
- Thus

$$1 \leq w_r(p) = \sum_{i=0}^{k-1} w_r(e_i)$$

$$= \sum_{i=0}^{k-1} [w(e_i) + r(v_{i+1}) - r(v_i)]$$

$$= w(p) + r(v_k) - r(v_0)$$

$$= w(p) + r(v) - r(u)$$

take the least $w(p)$ (tightest constraint) $r(u) - r(v) \leq w(u, v) - 1$

- Note: this is independent of the path from u to v, so we just need to apply it to u, v such that $D(u, v) > \propto$

Solving the Constraints

- All constraints in difference-of-2-variable form
- Related to longest/shortest path problem

Correlator: $\alpha = 7$

Legal:
$r(u) - r(v) \leq w(e)$

$$r(v_0) - r(v_1) \leq 2$$
$$r(v_1) - r(v_2) \leq 0$$
$$r(v_1) - r(v_3) \leq 0$$
$$r(v_2) - r(v_3) \leq 0$$
$$r(v_3) - r(v_0) \leq 0$$

Timing: D > 7
$r(u) - r(v) \leq w(u,v) - 1$

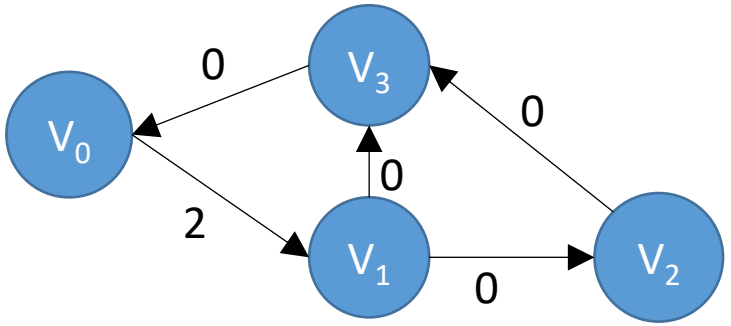
$$r(v_0) - r(v_3) \leq 1$$
$$r(v_1) - r(v_0) \leq -1$$
$$r(v_1) - r(v_3) \leq -1$$
$$r(v_2) - r(v_0) \leq -1$$
$$r(v_2) - r(v_1) \leq 1$$
$$r(v_2) - r(v_3) \leq -1$$
$$r(v_3) - r(v_1) \leq 1$$
$$r(v_3) - r(v_2) \leq 1$$

W

	V_0	V_1	V_2	V_3
V_0	0	2	2	2
V_1	0	0	0	0
V_2	0	2	0	0
V_3	0	2	2	0

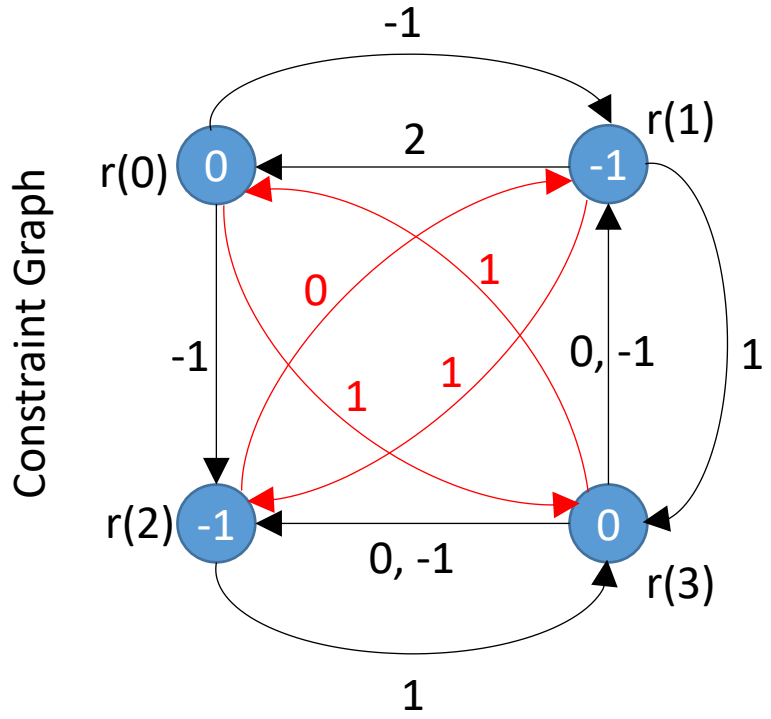
D

	V_0	V_1	V_2	V_3	
V_0	0	3	6	13	V_0
V_1	13	3	6	13	V_1
V_2	10	13	3	10	V_2
V_3	7	10	13	7	V_3



Solving the Constraints

- Do shortest path on constraint graph: ($O(|V|^3)$).
- A solution exists if and only if there exists no negative weighted cycle.



A Solution: $r(v_0) = r(v_3) = 0, r(v_1) = r(v_2) = -1$

Legal:
$r(u) - r(v) \leq w(e)$

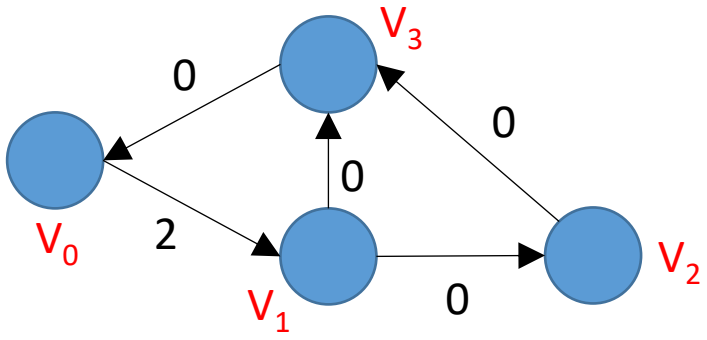
$r(v_0) - r(v_1) \leq 2$
$r(v_1) - r(v_2) \leq 0$
$r(v_1) - r(v_3) \leq 0$
$r(v_2) - r(v_3) \leq 0$
$r(v_3) - r(v_0) \leq 0$

Timing: D > 7
$r(u) - r(v) \leq w(u,v) - 1$

$r(v_0) - r(v_3) \leq 1$
$r(v_1) - r(v_0) \leq -1$
$r(v_1) - r(v_3) \leq -1$
$r(v_2) - r(v_0) \leq -1$
$r(v_2) - r(v_1) \leq 1$
$r(v_2) - r(v_3) \leq -1$
$r(v_3) - r(v_1) \leq 1$
$r(v_3) - r(v_2) \leq 1$

Retiming

- To find the minimum cycle time, do a binary search among the entries of the D matrix ($O(|V|^3 \log|V|)$)

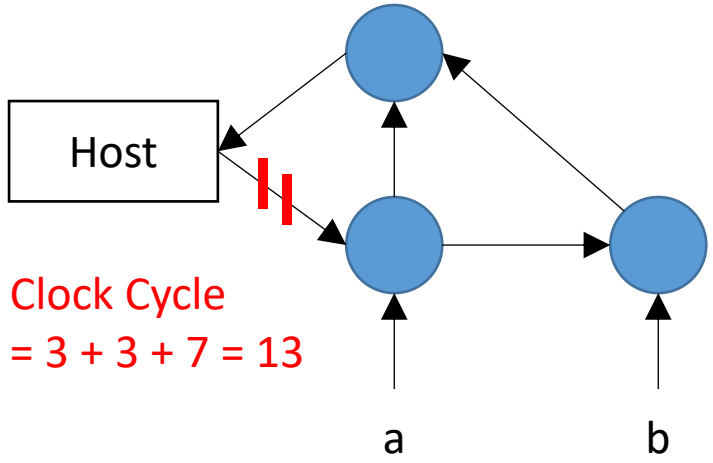


W

	v_0	v_1	v_2	v_3
v_0	0	2	2	2
v_1	0	0	0	0
v_2	0	2	0	0
v_3	0	2	2	0

D

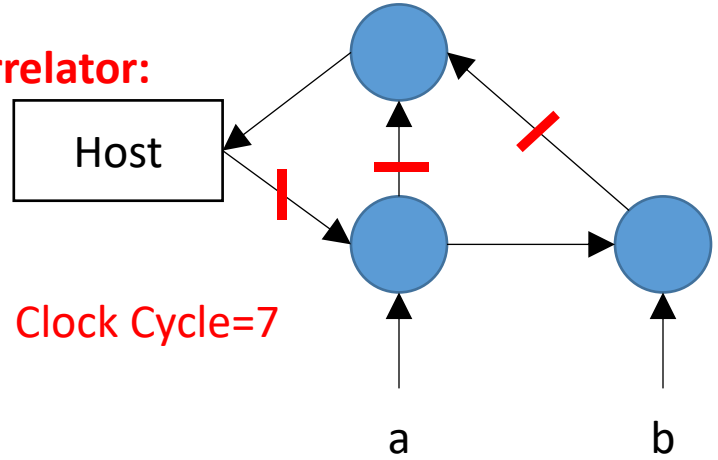
	v_0	v_1	v_2	v_3	
v_0	0	3	6	13	v_0
v_1	13	3	6	13	v_1
v_2	10	13	3	10	v_2
v_3	7	10	13	7	v_3



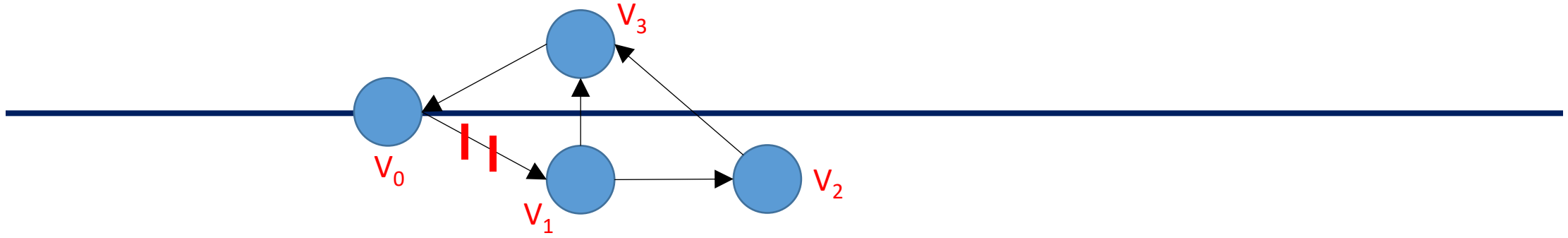
Clock Cycle
= 3 + 3 + 7 = 13

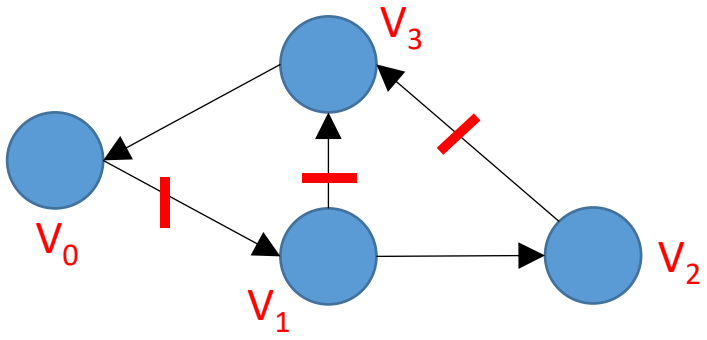
Retime

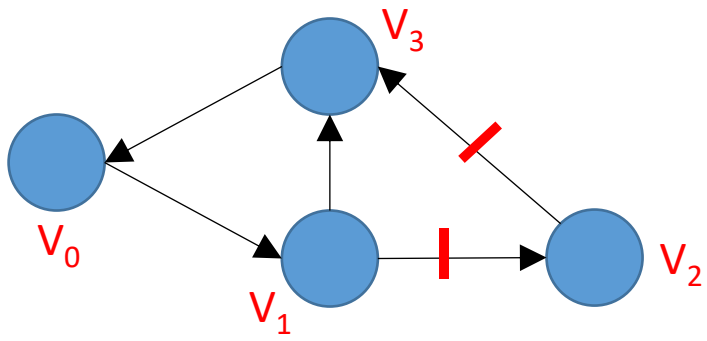
Retimed Correlator:

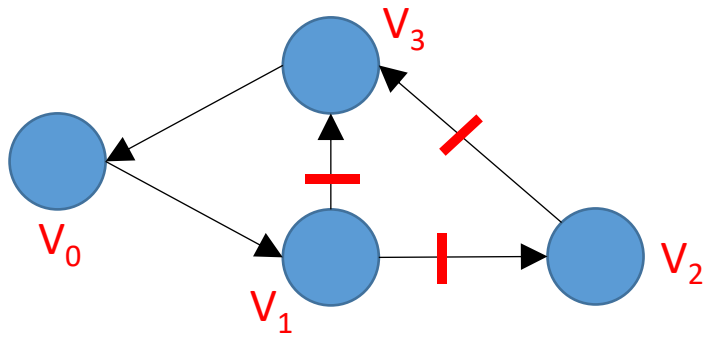


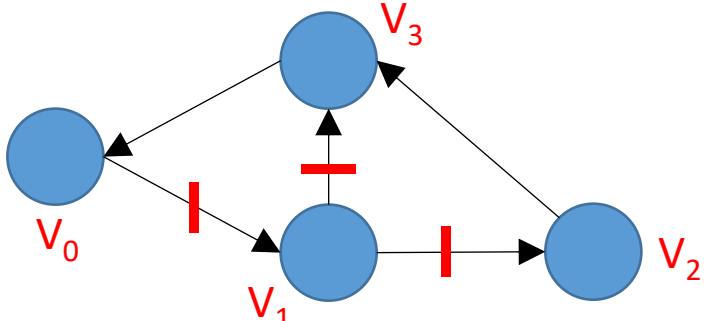
Clock Cycle=7

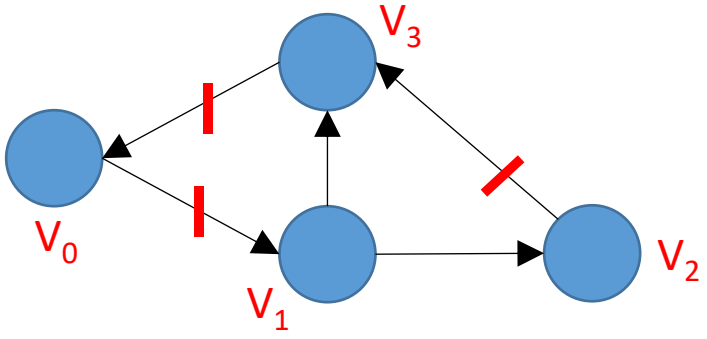

V3
V0
V2
V1

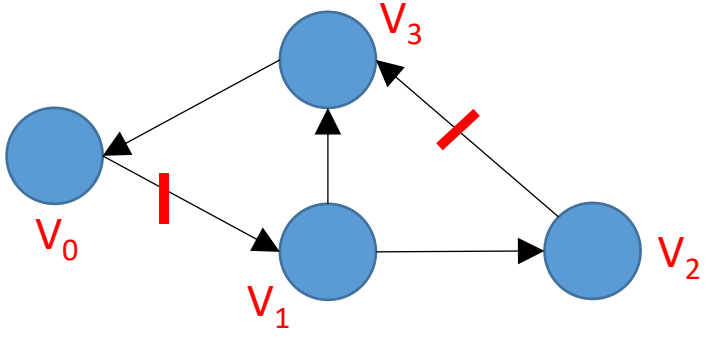

V3
V0
V2
V1


V3
V0
V2
V1


V3
V0
V2
V1


V3
V0
V2
V1


V3
V0
V2
V1


V3
V0
V2
V1

Retiming for Minimum Area (Minimum #Register)

- Goal: Minimize the number of registers used

$$
\begin{aligned}
minN_r &= \sum_{e \in E} w_r(e) \\
&= \sum_{e:u \to v} (w(e) + r(v) - r(u)) \\
&= \sum_{e \in E} w(e) + \sum_{e:u \to v} (r(v) - r(u)) \\
&= N + \sum_{u \to v} (r(v) - r(u)) \\
&= N + \sum_{v \in V} [r(v)(\#\ fanin(v) - \#\ fanout(v)] \\
&= N + \sum_{v \in V} a_V r(v)
\end{aligned}
$$

Where a_V is a constant for each node v

Minimum Registers - Formulation

- Minimize:

$$\sum_{v \in V} a_V r(v)$$

- Subject to: $w_r(e) = w(e) + r(v) - r(u) \geq 0$

Summary

- Retiming is a sequential optimization problem.
- Retiming can be used to solve
  - Minimize clock period problem
  - Minimize area (number of registers)
- Solving set of inequalities can be reduced to finding shortest path problem of graph
  - Polynomial time solvable
- Performance of retiming depends on the delay estimation of the combinational nodes
  - Combinational delay varies with target architecture, cell libraries, etc.
  - Accurate delay estimation is important

THANK YOU